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The DiskSat: A Two-Dimensional Containerized Satellite

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ABSTRACT

A key factor in the remarkable expansion of the CubeSat class of spacecraft over the past two decades is launch containerization. The container protects the launch vehicle and primary payload from issues that might arise from the CubeSat (which is essential for rideshare), and the standardized and highly-simplified launch interface reduces integration cost for the launch provider and development cost for the CubeSat builder. The downside of containerization is that the size of the contained satellites is rigidly limited. While there are available designs for larger dispensers and CubeSats, very few CubeSats larger than 6U have flown, and none have been larger than 16U. Future space missions will benefit from more power and RF aperture, beyond what can be provided by conventional CubeSats, even with complex deployables. We propose here the DiskSat, a containerized, large-aperture, quasi-two-dimensional satellite bus architecture.

A representative DiskSat structure is a composite flat panel, one meter in diameter and 2.5 cm thick, to which components are affixed in a flat pattern within the panel. The volume of the representative DiskSat is almost 20 liters, comparable to a hypothetical 20U CubeSat, while the structural mass can be less than 2.5 kg. The surface area of a single disk face is substantially larger than the total surface area of any conventional CubeSat, supporting over 200 W of peak solar power without the complexity of deployables, thereby improving mission assurance and reducing vehicle cost. Alternatively, a single fixed deployable panel can ensure that the vehicle has over 100 W orbit-average power while maintaining nadir pointing in any beta angle.

For launch, multiple DiskSats are stacked in a fully-enclosed container/dispenser using a simple mechanical interface, and are released individually once in orbit. Stacking of 20 or more DiskSats is possible in small launch vehicles, making it ideal for building large constellations of small satellites in multiple discrete orbital planes. The 1-m-diameter DiskSat was developed with the Rocket Lab Electron in mind; the concept can be extended to larger diameters (1.2 m for the Virgin LauncherOne, for example), or to other flat shapes (square for an ESPA port, for example), and to greater thicknesses if the mission requires it.

The DiskSat concept was developed as a cost-effective solution for a LEO constellation that required significant power and RF aperture. Since then we have explored the utility of the bus architecture for a broad range of missions including Earth observation and space science, among others. One particularly useful feature of the DiskSat is the high power-to-mass ratio, enabling high-delta-v electric propulsion missions, including deep-space applications. Another feature is the ability to fly in a low-drag orientation which, in combination with electric propulsion for drag makeup, enables flight at very low altitudes in LEO.

This paper will detail the design of the DiskSat and its dispenser, will explore the range of missions enabled by the DiskSat, and will describe current development activities in support of a DiskSat demonstration flight.

INTRODUCTION

In the past decade the cumulative number of CubeSats flown has increased from well under 100 to well over 1000.[1] A key factor in the remarkable expansion of the CubeSat class of spacecraft is launch containerization. The container protects the launch vehicle and primary payload from issues that might arise from the CubeSat, which is an essential feature that enables rideshare. The container also provides a standardized and highly-simplified launch interface that both reduces the integration cost for the launch provider and reduces the development cost for the CubeSat builder, ultimately enabling low-cost, mass-produced satellites. One consequence of containerization, however, is that the size of the contained satellites is rigidly limited. In the early days of CubeSats, this was of limited consequence; even though typical dispensers could

carry 3U satellites, until at least 2014 the majority of CubeSats flown were 1.5U or smaller. However, throughout the past decade CubeSats have increasingly been used in complex and operational missions, even while taking advantage of containerization. Particularly with the growth of commercial applications of CubeSats, often requiring more payload volume or power, interest in larger CubeSats accelerated. The first launch of a 6U CubeSat occurred in 2014 and by 2021 the cumulative number of 6U CubeSats flown has passed 100.[2] While there are available designs for larger dispensers and CubeSats, as of this writing fewer than 10 CubeSats larger than 6U have flown, and none have been larger than 16U.

As ever more sophisticated missions are proposed to take advantage of CubeSat containerization, they risk running into rather strict limits on surface area, which is needed for power, for heat dissipation, and for antennas. To some extent, these limits can be overcome using deployable components (solar arrays and antennas), but deployables can present challenges in themselves, and future space missions might benefit from yet more aperture, beyond what can reasonably be provided by conventional CubeSats, even with complex deployables. To overcome the aperture limitation, we propose here the DiskSat, a containerized, large-aperture, quasi-two-dimensional satellite bus architecture. In the DiskSat project, Aerospace is developing a new class of satellite that combines the benefits of containerization with the utility of large surface areas for power collection, thermal management, and RF apertures. Figure 1 shows a photograph of a mechanical prototype of the first generation of DiskSat, 1 m in diameter and 2.5 cm thick, seen next to the 3U ISARA spacecraft for size comparison. The DiskSat is intended to fill a gap in the mass-aperture space, particularly for constellations, that no rectangular prism satellite can practically fill. It is anticipated that the DiskSat will also support low-cost flight opportunities for student and other experimental flights that might require more surface area or volume than can reasonably be expected from a CubeSat, or that might simply benefit from the convenience of a more accessible and less constrained avionics and payload volume.

DiskSat Concept

Like the CubeSat, a DiskSat is defined first by geometry, but equally by the requirements of a standardized dispenser that delivers the DiskSat to orbit. Three principal characteristics define the satellite and dispenser combination:

1. The body of the DiskSat is made primarily of a flat, structural, plate in the form of a three dimensional solid with a thickness substantially smaller than other dimensions.

2. The satellites are designed to stack one upon another such that launch loads are taken up through the entire stack of satellites rather than having each satellite supported individually by a launch interface device.

3. Each satellite has an interface with the satellite dispenser that allows the satellites to be dispensed one at a time once the launch vehicle is in orbit.

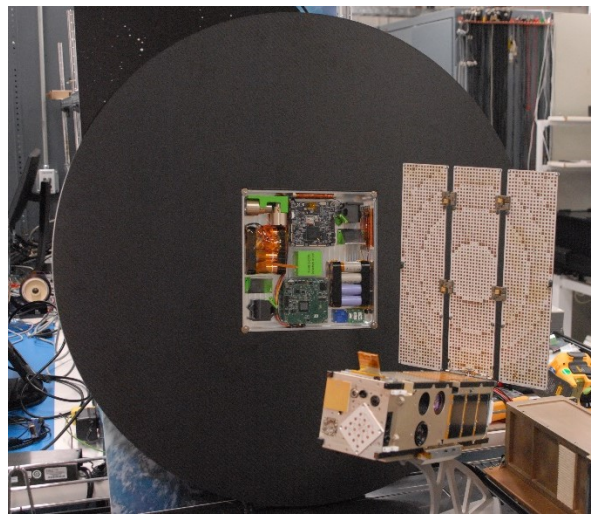

Figure 1. Photograph of a mechanical prototype DiskSat, 1 m in diameter and 2.5 cm thick, with a 3U CubeSat having a deployed antenna in the foreground for scale.

Also like the CubeSat, the DiskSat comes in a variety of sizes. However, DiskSat size limits are driven primarily by the available launch envelope. A representative DiskSat, sized to fit in the payload fairing of the Rocket Lab Electron,[3] has the shape of a circular disk, 1 m in diameter and 2.5 cm thick (see figure 1). Other standard sizes can be defined for other launch envelopes. For example, a disk with a diameter of 1.2 m is sized for the Virgin LauncherOne,[4] and a 1-m square "disk" would be compatible with an ESPA-Grande port[5] (see figure 2).

To minimize mass, the body of the DiskSat consists in large part of a composite sandwich panel fabricated in a manner similar to that used for deployable solar panels on larger spacecraft. For example, a panel may consist of an aluminum honeycomb core with graphite-epoxy face sheets, as illustrated in figure 3. Such structures are inherently low mass, with a typical area density under 3 kg/m^2, giving a structural mass of a 1-m-diameter disk of less than 2.5 kg. The total volume of the 1-m-diameter DiskSat is just under 20 liters, comparable in volume to a hypothetical 20U CubeSat. While this

entire volume could, in principle, be filled with avionics and payload, this would add substantially to the satellite mass, and it is anticipated that most DiskSats will fly with the majority of the internal volume unused; the goal of the DiskSat concept is to provide large apertures for power, thermal management, and RF aperture in a low-mass, containerized, satellite. The low-mass structure of the disk leads to a satellite that is not constrained by volume so much as by mass. For example, the standard Electron payload fairing has sufficient volume to easily contain a stack of 20 DiskSats 1 m in diameter and 2.5 cm thick. However, with the Electron payload capacity of ~230 kg, depending on orbit, and allowing 10 kg for the dispenser, the average mass of the DiskSats can be no more than 11 kg.

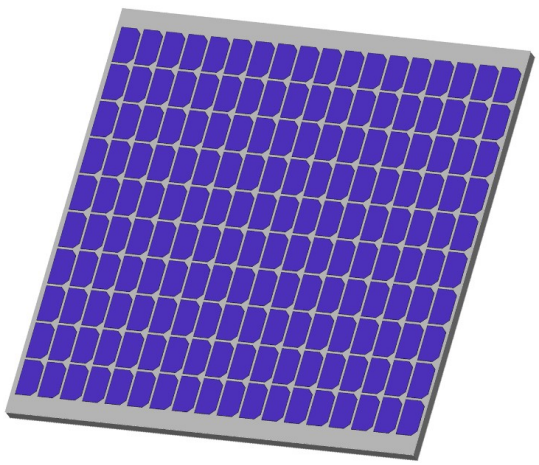

Figure 2. Perspective view of a 1-m square DiskSat, 2.5 cm thick.

The DiskSat approach has multiple advantages for a broad range of missions. The most notable, of course, is the aperture, which allows substantially higher power collection than is possible in a CubeSat. For example, the 1-m-diameter disk has enough surface area on one face to hold at least 200 W of solar cells without deployables. Similarly, the 1-m-diameter face can support a large-aperture phased-array or reflect-array antenna. Finally, there is enough surface area that portions can be specifically allocated to waste heat rejection without cutting substantially into the power-collection surface area.

A second key advantage of the DiskSat is ease of assembly. In typical CubeSats, the avionics and payloads have to be assembled in a compact three-dimensional package that can present significant integration challenges. In a DiskSat, the avionics and payload bay is more of a two-dimensional structure, where the components can be laid out over a large surface area, much in the manner of a flatsat. After integration, most components remain accessible, simplifying post-integration testing as well as removal and replacement of components if necessary.

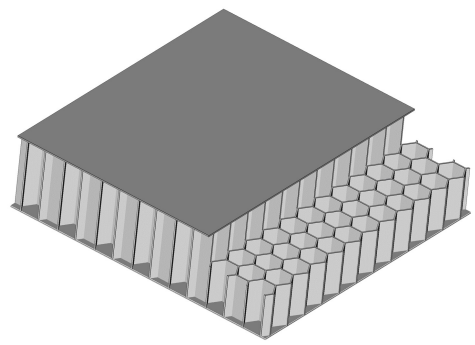

Figure 3. Cutaway view of typical composite sandwich structure showing top and bottom face sheets and an aluminum honeycomb core.

Of course, the new design does not come without challenges; the three most often noted are deployment, attitude control, and thermal management. Since the composite panel structure of a DiskSat is very rigid, the attitude-control challenge consists mainly of taking into account the larger moments of inertia, and driving the necessary reaction wheels. In a very simple analysis, assuming uniform mass density, the maximum moment of inertia (which is about the axis of cylindrical symmetry) of a 10-kg DiskSat will be about 30 times higher than the maximum moment of inertia of a 4-kg, 3U CubeSat. CubeSats, of course, are known for being particularly agile in attitude control, and it would require substantially larger and/or faster reaction wheels in a DiskSat to provide the same agility. However, for many missions agility is not really a requirement, and for those missions that do require agility the DiskSat should have adequate power to operate the larger/faster reaction wheels.

The thermal challenge will actually be twofold: first, minimizing thermal cycling due to the high surface-to-volume ratio; and second, disposing of large amounts of waste heat due to the high input power. On the plus side, for many applications most of the DiskSat internal volume will be purely structural and will not contain any temperature-sensitive components. In addition, the large surface area allows the spacecraft designer to dedicate specific surface areas to thermal management without sacrificing significant power-collection, RF-aperture, or sensor area. Finally, the inherently high power of the DiskSat allows the use of dedicated heaters to maintain temperature on critical components.

Launch And Deployment

As illustrated in figure 4, a DiskSat dispenser is configured to carry a stack of circular DiskSats in a cylindrical container (or square "DiskSats" in a square container), with the DiskSats stacked one upon another. The dispenser is intended to be the primary payload on a small expendable launch vehicle or on an ESPA port, and is intended to carry a stack of DiskSats large enough to make maximum use of the payload capacity. However, compatibility of the DiskSat dispenser with additional non-DiskSat payloads is also possible.

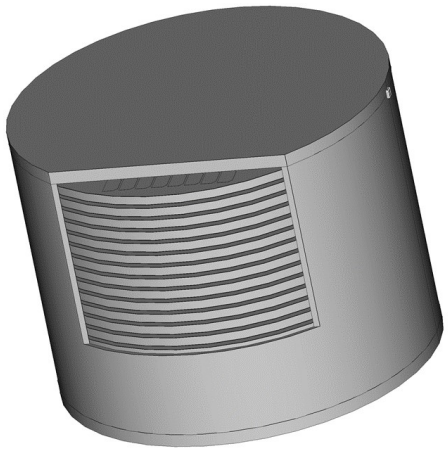

Figure 4. A 1-m cylindrical DiskSat dispenser with part of the lid and canister cut away to show some of the stack of 20 DiskSats in the dispenser.

Many of the requirements for the DiskSat dispenser are modeled on the requirements for the CubeSat dispenser, but there are requirements that are specific to the DiskSat. Requirements that closely follow the CubeSat model[6] include:

• The dispenser should protect the launch vehicle and primary payload (if any) from any mechanical, electrical or electromagnetic interference from the DiskSats.

• The mass of the dispenser should be kept to a minimum.

• The dispenser should incorporate a modular design that allows different numbers of DiskSats to be launched on any given mission.

• The resulting DiskSat standard should be easily manufactured without using exotic materials and expensive construction techniques

• The DiskSats should be released from the dispenser with minimal spin and with sufficient speed to ensure clean separation from the launch vehicle.

Requirements that are specific to the DiskSat dispenser include:

• The DiskSats should be released from the dispenser one at a time (either on command or with the use of a timer) to ensure low probability of recontact with the launch vehicle or with other DiskSats.

• While the dispenser containment volume may be unique for each launch vehicle, the dispenser mechanism should have a standard interface with the DiskSats that is independent of DiskSat size and common over a variety of launch vehicles.

The requirement that the DiskSats should be released one at a time is driven by the possibility of a DiskSat dispenser containing large numbers of satellites in a single launch. With CubeSats, the original P-POD was designed to contain three 1U satellites, one 3U satellite, or some other combination of satellites totaling 3U. The CubeSat dispenser releases the CubeSats simply by opening the cover, and will release all contained satellites simultaneously. With no more than three satellites contained in the dispenser, the probability of satellite-to-satellite contact after deployment is small. The DiskSat dispenser, in contrast, might contain 20 or more satellites in a single launch. If these were all released simultaneously, the probability of satellite-to-satellite contact increases substantially, particularly if the DiskSats in the stack are not all of the same mass, or have some asymmetry in their mass distributions.

The requirement for individual release of the DiskSats implies a certain level of complexity in a mechanism that restrains each DiskSat individually prior to release and then releases each DiskSat individually in turn. In typical space launch systems carrying multiple payloads, each satellite is attached to the launch vehicle by an interface mechanism that both carries the launch loads and operates in space on command or schedule to release the satellite. This requires a mechanism that is both robust (to carry the launch loads) and complex (to release the satellite at the proper time). To minimize the mass of the DiskSat dispenser, we have chosen an alternative approach where the launch loads are carried separately from the dispensing mechanism. Specifically, the DiskSats are attached to the dispensing mechanism with sufficient compliance to ensure that launch loads are, at most, weakly transmitted between the DiskSat and the dispensing mechanism. At the same time, the DiskSats are stacked upon one another and held in the launch stack by the container such that the lateral launch loads are carried by the container walls and the vertical launch loads are carried through the stack between the floor and cover of the container.

The containment is, in fact, similar to the way multiple CubeSats are contained in a CubeSat dispenser. In the CubeSat standard,[7] a 3U CubeSat has four rails on the long edges that are required to be 340.5 mm long, a dimension that matches the internal length of a 3U dispenser. When the lid of the dispenser is closed, the CubeSat is constrained in the axial direction by forces on the ends of the rails. When a CubeSat dispenser contains three 1U CubeSats, each of the CubeSats has rails that are 113.5 mm long so that, when stacked in the dispenser, the combined length of the rails is 340.5 mm. When the dispenser door is closed, the three CubeSats are constrained in the axial direction by forces on the ends of the rails in contact with the dispenser and by the CubeSat-to-CubeSat contact, which occurs only at the rail ends.

In principle, a stack of DiskSats could be constrained in the axial direction in the same manner, using either a shoulder on the perimeter of the disk, or a defined set of perimeter contact points analogous to the CubeSat rails, to carry the vertical load in the stack. In practice, however, a DiskSat one meter in diameter, even when made with a material as rigid as a graphite/epoxy sandwich structure, will flex under launch loads if held only at the perimeter, and vibration-induced contact between adjacent disks in the stack could be problematic. To counter this, the DiskSat standard will include additional contact points distributed across the surface (see below). The floor and lid of the container will have matching contact points and the vertical load in the stack will be carried through these contact points in the same way that the vertical load in a CubeSat stack is carried through the rails.

While the vertical loads on the stack are carried through the stack, necessitating satellite-to-satellite contact, each satellite in the stack is expected to have a diameter equal to the diameter of the stack, allowing direct contact between the perimeter of each disk and the wall of the container, and allowing lateral loads to be carried individually by each satellite, in the same manner that lateral launch loads in CubeSats are carried by contact between the rails and the sides of the CubeSat dispenser.

Once the dispenser reaches orbit, the lid can be opened, removing the launch-load constraint, and the DiskSats are now held in the stack only by the release mechanism. Since the release mechanism does not have to carry the launch loads, and only needs to control the release of the DiskSats in zero gravity, the mechanism need only be robust enough to carry its own weight against launch loads. The details of the release mechanism are still being developed, however it is anticipated that it will lift each disk successively from the top of the stack and release it with sufficient speed to ensure clear separation from the launch vehicle. The interval between successive disk releases can be controlled either by a timer or by individual commands for each disk.

DiskSat

Again in analogy to the CubeSat standard, the DiskSat dispenser will impose certain requirements on the satellite:

• Basic dimensional and mass requirements ensure the DiskSat fits properly in a stack and in the deployer.

• Contact points or standoffs on the top and bottom faces of the DiskSat are required to provide separation between DiskSats.

• A kill switch is required to ensure the satellite is powered off while in the deployer. Along with the kill switch there is also a requirement for a “remove before flight” pin, to deactivate the DiskSat while it is being loaded into the deployer.

• While the CubeSat standard requires that deployable components (solar panels, covers, antennas, etc.) not be released until 30 minutes after the CubeSat leaves the CubeSat dispenser, that requirement has occasionally been relaxed in recent years and some (but not all) dispensers are configured such that deployable components are restrained only by the dispenser walls and therefore deploy immediately after CubeSat release. Because of the increased complexity of the satellite release process in the DiskSat, it is expected that a restriction on early release of deployables will need to be uniformly enforced for DiskSats.

• The DiskSat will have to include an interface with the dispenser release mechanism. While that mechanism has yet to be finalized, the design approach we are taking is to place as little complexity as possible on the DiskSat; it is expected that the DiskSat's interface with the dispenser will be a simple drop-in component that can be attached to the DiskSat late in the manufacturing process; this component will not be required to carry the mass of the DiskSat against launch loads.

Within these requirements, there is a fair amount of flexibility in how the DiskSat is built and what it can carry.

DiskSat Structure

In general, the DiskSat is in the shape of a plate - a three-dimensional solid with a thickness that is small in comparison to its other dimensions. As shown in figure 1, the plate is circular and the satellite is in the form of

a disk. In alternative configurations, the plate may take on other geometric forms, such as a square, a hexagon, or an octagon, to name a few, or the shape of the plate may be irregular, depending on the application. The key geometric feature of the DiskSat is that the in-plane dimensions are substantially larger than the dimension perpendicular to the plane.

The initial motivation for the DiskSat geometry was to provide a large surface area in a satellite with a mass comparable to a typical CubeSat. This is most easily accomplished by basing the DiskSat structure on a composite sandwich panel, a structural form often used, for example, for deployable solar panels on larger spacecraft. Such sandwich panels typically comprise two face sheets bonded to a core material, where the face sheets might be aluminum or composite materials such as glass/epoxy or graphite/epoxy and the core material might be either a honeycomb (as illustrated in figure 3), or a foam material, either of which is designed to provide structural rigidity to the sandwich while minimizing mass.

Contact Points

The CubeSat Design Specification limits the dimensions of any components on the end faces of the CubeSat relative to the height of the rails to prevent contact other than at the rails. For satellites as small and rigid as CubeSats, this is generally sufficient to ensure that they do not make contact with one another at other points. For stacked satellites in the form of plates, limiting contact between the satellites to the satellite edges might not provide a robust separation between the satellites across the gap. Because of this, the DiskSat specification will include an array of contact points distributed in a defined pattern across the top and bottom faces of the DiskSat such that a DiskSat in a stack will make contact with the DiskSats above and below it only at the defined contact points. The points will protrude from the surface of the DiskSat sufficiently to ensure that launch-induced deformation of the DiskSat between contact points will not be large enough to allow adjacent DiskSats to touch anywhere other than at the defined contact points.

Figure 5 shows two views of a stack of five satellites illustrating the use of load-bearing pins to provide the contact points. In the perspective view, load-bearing pins are seen protruding from the top surface of the top DiskSat in the stack. The placement of the pins in the array is designed to mesh with an array of solar cells on the same face. The side view shows how the pins in adjacent satellites align, and how the total length of the pins defines the center-to-center spacing of the satellites in the stack. When the stack is installed in a DiskSat dispenser for launch, the pins will align with matching contact points on the floor and lid of the dispenser. In this manner, the array of contact points allows for the launch loads to be carried through stack without surface-to-surface contact between the satellites in the stack. Since the stability of the stack depends on proper alignment of the contact points, the DiskSat specification will include requirements for precise lengths and placement of the pins, in a manner similar to the requirements for CubeSat rails.

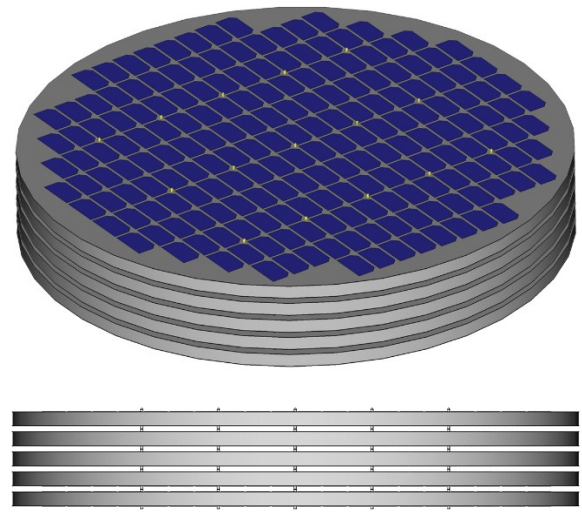

Figure 5. Two views of a stack of five DiskSats showing the array of load-bearing pins used to carry vertical loads and maintain separation between adjacent DiskSats in the stack.

Distribution Of Components

Some satellite components, such as solar cells, radiators, and antennas, need exposure directly to space. Other components, such as electronics, reaction wheels, torque rods, and batteries, are typically contained in the internal volume of spacecraft. In a DiskSat, the ratio of surface area to volume is large compared to a typical spacecraft, and the placement of solar cells and antennas on the surface is straight forward.

Surface Components

Surface components, such as solar cells or antennas, can be mounted directly to the face sheet of a DiskSat. In the case of solar cells, for example, the face of a DiskSat may incorporate mount points as well as the wiring to route the power from the solar cells to the power system of the DiskSat. If other considerations (e.g., structural, thermal, etc.) allow, this might even be facilitated by using a printed circuit board as the face sheet. Solar cells might be distributed on one or both face sheets. In many applications, a DiskSat might orbit with one face always directed toward Earth (nadir)

while the opposite face (anti-nadir) is always pointing away from the Earth. In low Earth orbits, the nadir face is rarely exposed to the sun, so the solar cells on the nadir face are rarely illuminated, and therefore, are of limited utility. On the other hand, satellites might not always be under attitude control (e.g., when first launched, or during operational anomalies), and it might therefore be beneficial to have at least a few solar cells on the nadir face to provide a minimal amount of power should the satellite find itself in an orientation where the anti-nadir face is always pointing away from the sun.

Internal Components

With a thickness of 2.5 cm, a disk-shaped satellite one meter in diameter has a volume of 19.6 liters, allowing more than adequate volume within the satellite for all components not needing to be on one of the faces. However, although the overall volume is large, one of the dimensions is small so the components need to be laid out differently than in traditional satellites. One option is to provide a set of cavities in the core of the sandwich structure to make space for satellite components. Figure 6 illustrates the placement of various non-structural components, including avionics and payload, between the face sheets in cavities or voids in the core material. Such placement may be facilitated with mount points provided on the face sheets for the various components, and wiring harnesses running between the voids.

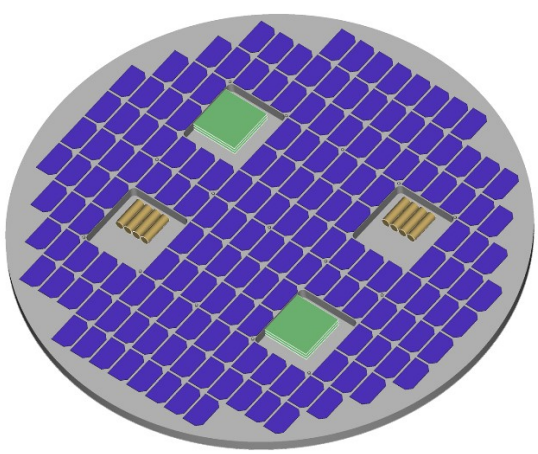

Figure 6. Perspective view of a 1-m DiskSat with four voids in the core material for holding non-structural satellite components. (The covers over the voids are hidden to make their contents visible.)

In an alternative configuration, the sandwich panel may include one or more gaps, through the core material and through one or both face sheets, into which a chassis box containing non-structural satellite components may be inserted, as illustrated in figure 1, where the chassis box cover has been removed for clarity. Note that the chassis box also incorporates a structural element in the form of a load pin supporting top and bottom contact points to prevent there being any gaps in the pattern of contact points. Since the core material has a low density, the placement of components determines the mass properties of the DiskSat, so it might be preferable to place the chassis box near the physical center of the satellite to minimize the moments of inertia.

Oversize Components

Since the DiskSat is in the form of a plate, with the intention to create a stack of DiskSats for launch, the thickness of each DiskSat is limited by the number of satellites intended to be stacked and the height of the available launch volume. Assume, for example, that a deployer payload volume is 1 m in diameter and 1 m high. In this example, a stack of 25 DiskSats may be accommodated if each disk is no more than 4 cm thick. In practice, many satellites do not need to be that thick; as noted, a disk 1 m in diameter and 2.5 cm thick has a volume of about 20 liters. By way of comparison, over 1000 CubeSats launched to date have had volumes of 3.5 liters or smaller. In practice, the thickness of a satellite plate need be no more than required to contain all necessary satellite components. Since many satellite components and payloads are primarily electronic in nature and the electronics can be laid out in flat arrays, a thickness of 2.5 cm might often be more than adequate. In general, any satellite component having at least one of its three dimensions smaller than 2.5 cm can be incorporated in a satellite that is 2.5 cm thick. In applications that require components having no dimensions smaller than 2.5 cm, the thickness of the disk may be increased. For example, a disk having a thickness of 10 cm would allow the incorporation of any component that might fit in any CubeSat that is 6U or smaller.

Although it is expected that most satellite components could be installed in the volume of a relatively thin plate of the satellite, there might be components that do not fit because the dimensions are larger than the thickness of the satellite plate, or because a specific orientation relative to the satellite plane is required. An alternative configuration, that allows thick components while minimizing the average thickness of the DiskSat, is to let certain components protrude beyond the surface of the satellite plate while providing matching cutouts in adjacent satellites in the stack to allow close stacking of the satellites. Figure 7 illustrates a DiskSat having an electric propulsion module that is too large to fit in within the 2.5-cm thickness of the DiskSat plate. The DiskSat in this illustration also has a cutout in the disk 180 degrees opposite the thruster. The cutout in this configuration makes it possible to have a close stack of

identical or nearly identical satellites provided that cutouts are spaced appropriately and the satellites are clocked in the stack, allowing for a protruding element of one satellite to align with a cutout in an adjacent satellite. Figure 8 illustrates how successive satellites are rotated 180 degrees relative to the prior satellite for stacking, repeating the pattern as often as necessary to build the full stack. A satellite could also incorporate a protruding element not at the perimeter of the disk, provided that the corresponding gap, offset 180 degrees from the protruding element, is also not at the perimeter of the disk. Note, however, that it is not possible to stack identical satellites with protruding elements at the center of symmetry. A satellite could also incorporate multiple protruding elements of varying sizes provided the corresponding gaps map correctly to the protruding elements with a defined rotation.

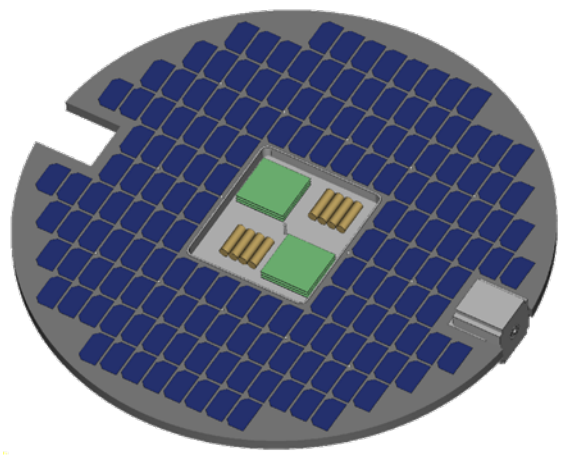

Figure 7. Perspective view of a 1-m DiskSat with an oversize electric propulsion module and a cutout to facilitate stacking.

External Features

Although the DiskSat has a very large surface area for its volume or mass, there could be applications where the area of the disk itself is insufficient and deployable structures would be needed. In this case, the planar structure of the satellite would facilitate large-area deployable components. For example, as illustrated in figure 9, a deployable solar panel, also in the shape of a disk, is connected via a hinge to the DiskSat. Although not easily visible in figure 9, both the DiskSat and the deployable panel include load pins to provide contact points with adjacent satellites in the stack as well as contact points between the DiskSat and the panel. Figure 9 shows the panel in the deployed position while figure 10 shows a closeup of the load pins with the panel stowed in the launch configuration. Specifically, the load pins in the disk protrude below the disk a distance sufficient to ensure standoff between the DiskSat and a satellite that might be stacked below it during launch. The load pins in the disk also protrude above the disk a distance approximately half the width of the gap between the disk and the panel. Similarly, the load pins in the panel protrude below the panel approximately half the width of the gap such that two load pins meet at a contact point and maintain a fixed separation between the disk and the panel. Finally, the load pins in the panel also protrude above the panel a distance sufficient to ensure standoff between the panel and a satellite that might be stacked above it.

For other applications, a non-deployable stand-off panel might be useful, for example, for thermal control or for isolation of an RF antenna. In this case, the load pins will extend through and be connected to both the disk and the panel, maintaining the separation between them. As in the case with a deployable panel, the load pins will protrude beyond the surface of both the disk and the panel sufficiently to provide contact points with adjacent satellites that might be stacked above or below the satellite during launch.

Manner Of Flying

Satellites in space generally experience very little drag because the atmospheric density is exceedingly low. However, in low Earth orbit (LEO), atmospheric density increases as orbit altitude decreases, and atmospheric drag becomes significant, limiting the orbit lifetime for most satellites orbiting below about 800 km altitude. The effect of atmospheric drag is inversely

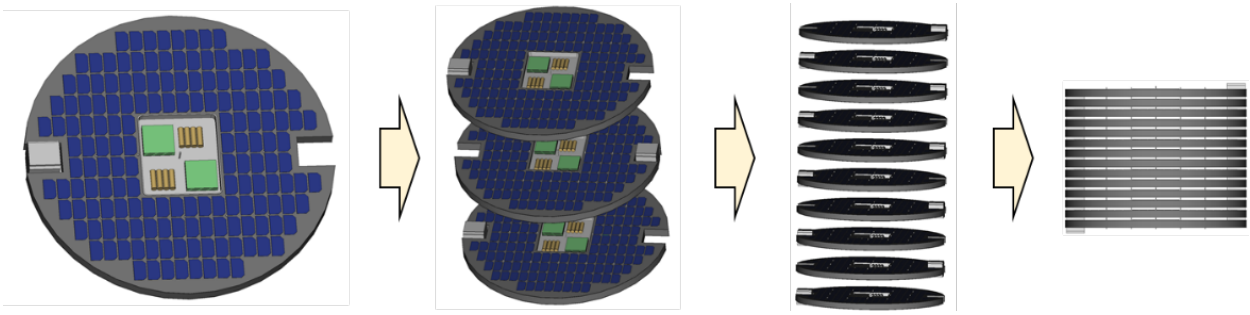

Figure 8. Diagram illustrating how successive satellites are clocked for stacking.

proportional to the ballistic coefficient (the ratio of mass to cross sectional drag area); satellites with a high ballistic coefficient are less affected by atmospheric drag than satellites with a low ballistic coefficient. For a flat plate in LEO, since the mass is constant, but the cross-sectional drag area is a function of flight attitude, the flight attitude has a significant impact on the ballistic coefficient, and therefore atmospheric drag. In general, unless the satellite has passed its end of life, it is preferable to fly the satellite in a low-drag orientation. For example, if one face of the plate includes a payload with a field of view normal to the face that needs to be pointed at nadir, then the spacecraft will normally be flown in a low-drag orientation. If, on the other hand, the satellite contains a payload on the perimeter with a field of view in the plane of the disk that needs to be pointed at nadir, then the satellite would preferably be flown in a low-drag attitude where the plane of the satellite disk is in the plane of the orbit and the surface normal of the satellite disk is perpendicular to both the velocity vector and the vector pointing at nadir.

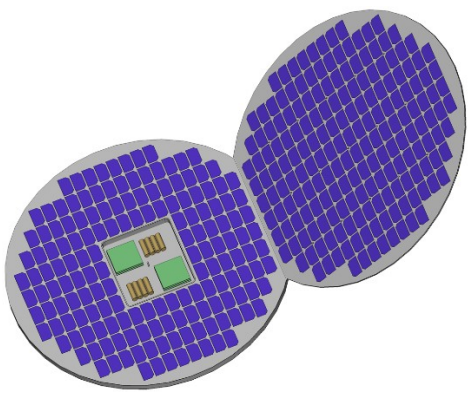

Figure 9. Perspective view of a 1-m DiskSat with a single deployed solar panel.

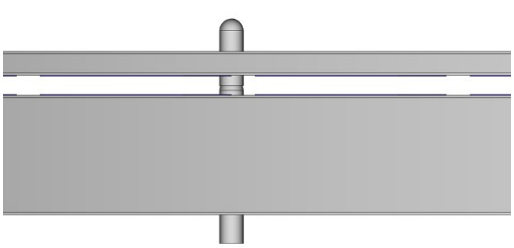

Figure 10. Close-up view of a single set of load pins between a DiskSat and its deployable solar panel when in the non-deployed configuration for launch.

If, on the other hand, the satellite has reached the end of its useful life, but still has an operational attitude-control system, the satellite can be flown in a high-drag mode with the plate-normal vector parallel to velocity vector, resulting in a very high drag. For example, a disk 1 m in diameter and 2.5 cm thick will have a cross sectional area of 250 cm^2 in a low-drag orientation, and a cross sectional area of 7850 cm^2 in a high-drag orientation. Depending on initial altitude, the substantial increase in drag area might be used to quickly deorbit a plate satellite at end of life. Even in the case where the satellite fails completely, leaving it without attitude control, it can be expected that the satellite will tumble, giving it an orbit-average drag area not as high as an intentional high-drag orientation, but still substantially higher than an intentional low-drag orientation, leading to efficient disposal of non-functional satellites in LEO.

For a satellite flying with a deployable solar panel, as previously illustrated in figure 9, it is still possible to fly in a low-drag orientation, but the choice of orientation is more restricted. Specifically, the drag can be minimized if the velocity vector is parallel to the axis of the panel hinge. This is illustrated in figure 11, where the nadir face of the satellite disk is pointed along the Earth-nadir vector (to support, for example, a payload in the disk face that needs to be pointed at Earth). The solar panel is deployed about 30 degrees above the plane of the disk such that a vector normal to the surface of the solar panel, and a vector normal to disk, are both perpendicular to the velocity vector (which is into the page in figure 11). This configuration would also be useful for maximizing available solar power as the plane of the orbit precesses relative to the solar direction. For example, when the beta angle (the angle between the plane of the orbit and a vector to or from the sun) is low, the solar cells on the anti-nadir face of the disk and the cells on the deployed panel are illuminated over half of each orbit, with the angle of illumination varying between zero and 90 degrees (relative to the plane of the disk), and the orbit average available power is about one third of the peak power. When the beta angle is high (90 degrees), the solar cells on the anti-nadir face of the disk are not illuminated during any part of the orbit, but the cells on the deployed panel are illuminated continuously (since the satellite never goes through eclipse when the beta angle is high). Because of the angle of illumination, the power collected by the cells on the deployed panel is reduced by about a factor of two relative to the peak possible power with direct normal illumination, but the power is available throughout the orbit. As such the orbit average power available to the satellite is about the same at either high or low beta angles. Intermediate beta angles will provide higher peak power to cells

mounted on the deployed panel and some power to cells mounted on the disk. The deployment angle of the panel can be selected to tune power availability for the intended orbit.

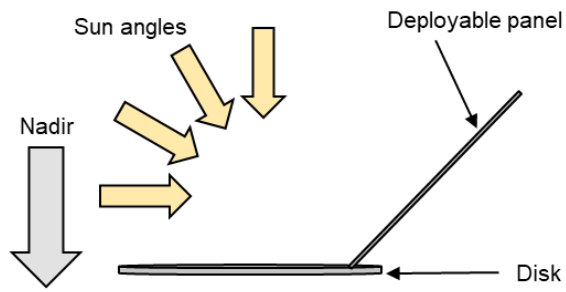

Figure 11. Side view showing the low-drag orientation of a satellite with a single deployed solar panel. The velocity vector is perpendicular to the page, the nadir direction is as indicated, and the beta angle can range between 0 and 90 degrees as the orbit precesses.

The potential utility of this approach is illustrated in figure 12, where published data for several commercially-available CubeSat and SmallSat buses is used to compare orbit-average power to bus mass. Traditional satellites with relatively simple deployable solar panels can be grouped on a trend line. One available 12U bus falls well above this trend line by using two 17-fold deployable solar panels. Several DiskSats with various diameters are also included in the plot, some with and some without a single fixed deployable solar panel. Because of its large surface area and low mass, the DiskSats, whether with or without a deployable solar pane, fall on an entirely new trend line.

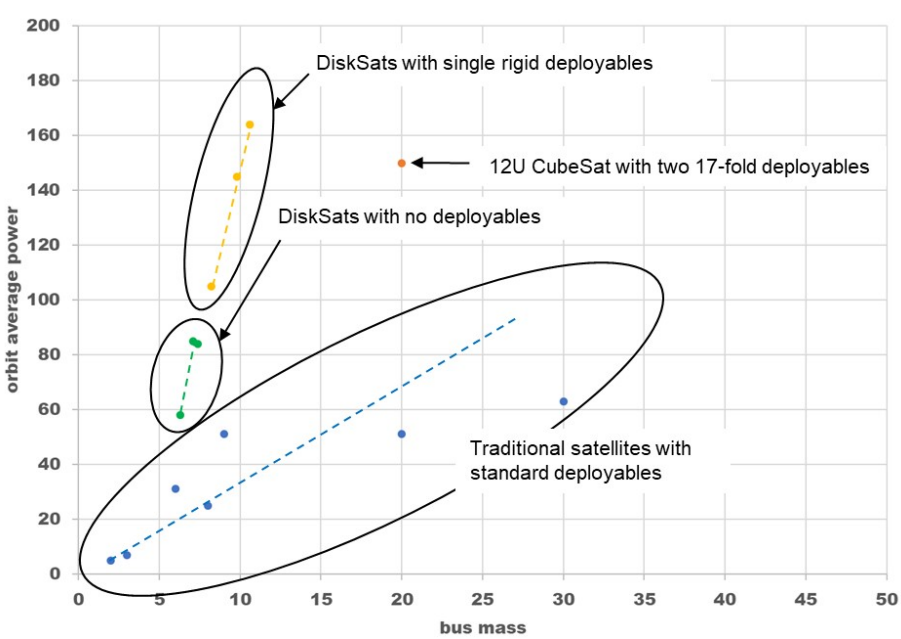

Figure 12. Plot of spacecraft orbit-average power as a function of spacecraft mass for representative small satellites and various DiskSats. The data points are approximations based on published specifications for selected spacecraft.

DISKSAT APPLICATIONS

Although it is possible to envision a DiskSat taking on almost any mission currently envisioned for CubeSats, two key characteristics of the DiskSat suggest specific mission applications; the high power-to-mass ratio, and the stackability, the latter of which allows for efficient use of launch capacity in small launch vehicles.

Constellations

There have been many proposals for satellite constellations (some of which are currently under development) that involve hundreds to thousands of satellites, enough to fill the payload capacity of any existing launch vehicle even if the satellites can be reduced in mass to a few kg each. However, large constellations typically require multiple satellites be deployed in each of multiple orbital planes. Since it is challenging for multiple satellites launched on a single launch vehicle to be deployed in more than one orbital plane, but relatively easy for multiple satellites on a single launch to be distributed about a single plane, the number of launches required to deploy a constellation typically will be of the same order as the number of orbital planes in the constellation. While it has become the norm to fly small satellites as rideshare as much as possible, it can be problematic trying to build a well-structured constellation using rideshare since the secondary payload is generally left in the same orbital plane as the primary. On the other hand, if the number of satellites that can be used in a single orbital plane is large enough to fill the payload capacity of an entire launch vehicle, then it might make sense for a satellite operator to purchase entire launch vehicles instead of going as rideshare payloads. As such, the key metric in determining the compatibility of a launch vehicle and constellation design is the comparison of the payload capacity and the

combined mass of all satellites in a single orbital plane. For example, a constellation that uses 25 satellites per plane, with each satellite having a mass of 200 kg, will require 5000 kg delivered to each orbital plane. Such a constellation could make efficient use of a launch vehicle with 5000 kg capacity. On the other hand, a constellation that uses 50 satellites per plane, where each satellite is only 5 kg, would require only 250 kg per plane, and would not be able to justify using a typical large launch vehicle.

Launch vehicles designed for (and appropriately priced for) smaller payloads are becoming more generally available, making it possible to think about dedicated launches for small-satellite constellations requiring only a few hundred kg per orbital plane. The DiskSat concept was initially developed during a concept study of a mission that required a well-structured constellation of several hundred CubeSat-class satellites in LEO, and was designed specifically for this application. The geometry of the satellite is intended to provide maximum power and aperture with minimum mass while taking maximum advantage of both payload volume and mass capacity of small launch vehicles. Although the concept was initially developed for a specific constellation mission, it is more broadly applicable to most missions that envision a well-structured constellation.

Agile Satellites

As noted above, the DiskSat might not be as agile as a typical CubeSat in the sense of agile attitude control. On the other hand, agility can also refer to orbit changes, and the high power-to-weight ratio of the DiskSat makes it an ideal candidate for high-delta-v missions using electric propulsion. Consider, for example, a DiskSat such as that shown in figure 7, where the electric thruster might be, for example, an Enpulsion Micro R3.[8] This unit draws up to 120 W of power, depending on operating parameters, carries 1300 g of propellant, and operates at a maximum specific impulse of 6000 seconds. If the DiskSat mass, including the propulsion unit, is assumed to be 11 kg, then the total delta-v possible is well over 4000 m/s, which is enough, for example, to fly from GEO to almost anywhere in cis-lunar space, including lunar orbit. On the other end of the spectrum, the combination of high power, continuous propulsion, and low drag makes it possible to fly a DiskSat at very low altitudes. Depending on solar activity, the drag force on a DiskSat flying in low-drag orientation in a circular orbit at 200 to 250 km is low enough to allow drag makeup using an Enpulsion Nano.[9,10] This unit carries 220 g of propellant, enough to provide drag makeup for over a year at these low altitudes. This opens a new mission space for low-altitude imaging, radar, or other Earth-observation missions.

CURRENT STATUS

Aerospace has an ongoing project to understand and mitigate potential risks of the DiskSat concept and to develop a detailed plan for three-satellite LEO demonstration mission to validate the launch dispenser and to validate the basic satellite design. While the satellite design is novel, most of the avionics required for a demonstration mission can be adapted directly from the avionics already in use in the AeroCube series of CubeSat flights developed at Aerospace over the past 15 years. The key challenge with the DiskSat is expected to be developing and demonstrating a reliable mechanism for launch and release. Like the CubeSat, our intention is to develop a system that is universal, to the maximum extent possible, for all DiskSats across a broad range of launch vehicles. Current work is focused on developing a reliable release mechanism. Various options are being considered, and prototype designs have been developed. Fabrication and initial testing is expected to be completed in 2021, with a potential for a first flight as early as late 2022.

SUMMARY AND CONCLUSION

The DiskSat is a containerized, large-area, quasi-two-dimensional satellite bus architecture, designed to bring the benefits of containerization to a low-mass, high-power satellite bus able to support a broad range of missions that would be very challenging within the CubeSat form factor. The key distinguishing features of the DiskSat are the flat-panel construction and stackability. A baseline DiskSat, sized for the Rocket Lab Electron, is 1 m in diameter and 2.5 cm thick, and can support orbit-average power levels over 100 W with a bus mass under 10 kg. The geometry of the DiskSat facilitates efficient packaging for launch, with the option to stack 20 or more at a time in a single launch, limited by the mass capacity of the launch vehicle rather than the payload fairing volume. Variations on the basic DiskSat geometry enable efficient use of other small dedicated launch vehicles or rideshare on an ESPA port. The DiskSat form factor is ideally suited to applications requiring large, well-structured, constellations of small satellites in LEO, and might be a cost-effective solution for any small satellite that requires significant power and/or RF aperture. Combining the DiskSat with electric propulsion enables high-delta-v missions such as from GEO to lunar orbit, or sustained flight at very low altitudes.

Aerospace is currently working on developing a demonstration mission, with initial focus on risk reduction in the deployer, the power system, and the

attitude-control system, with to goal eventually of developing a universal DiskSat standard in the mold of the CubeSat standard.

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